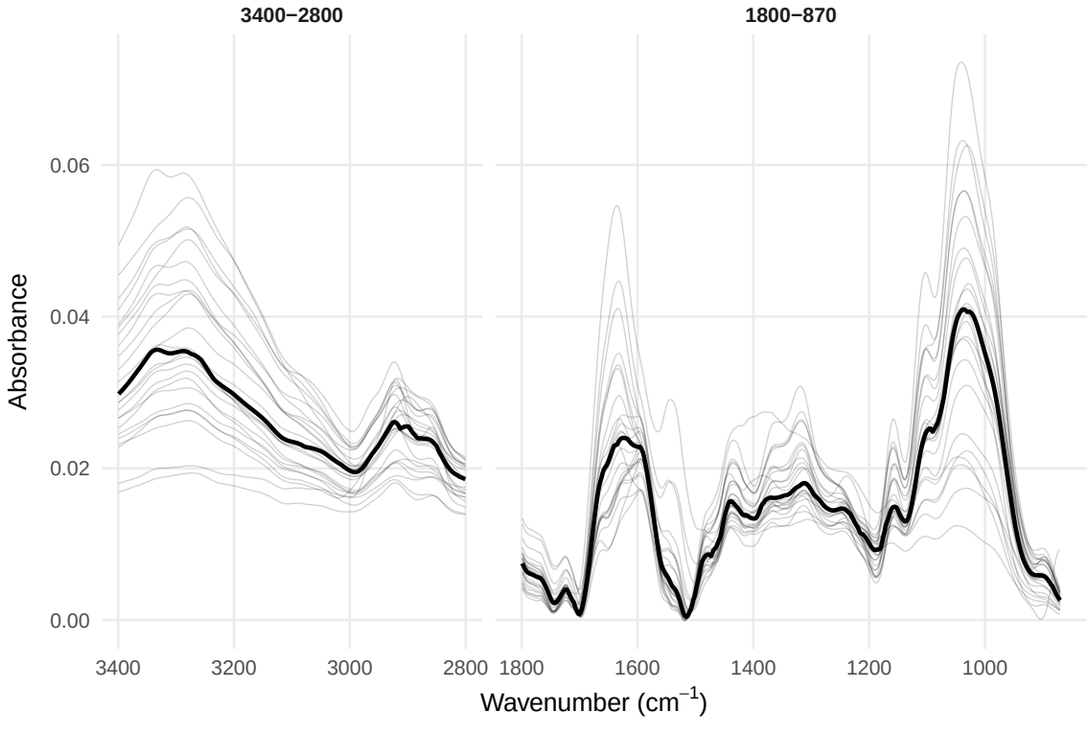
A

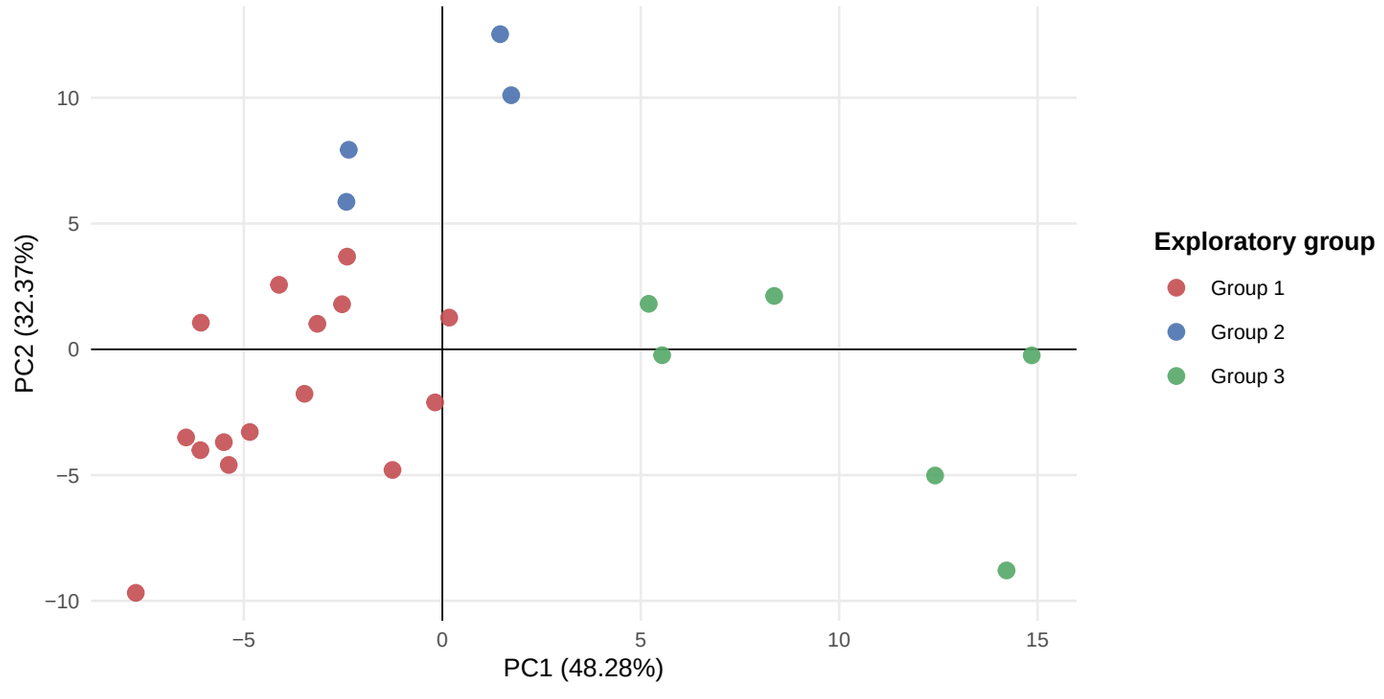
Participant−level absorbance spectra

Thin lines: individual measurements; thick line: cohort median



B

Exploratory three−group representation



C

Continuous PC1−PC3 eigenvector loading profiles

Raw mathematical PCA signs are retained; no post hoc orientation is applied

